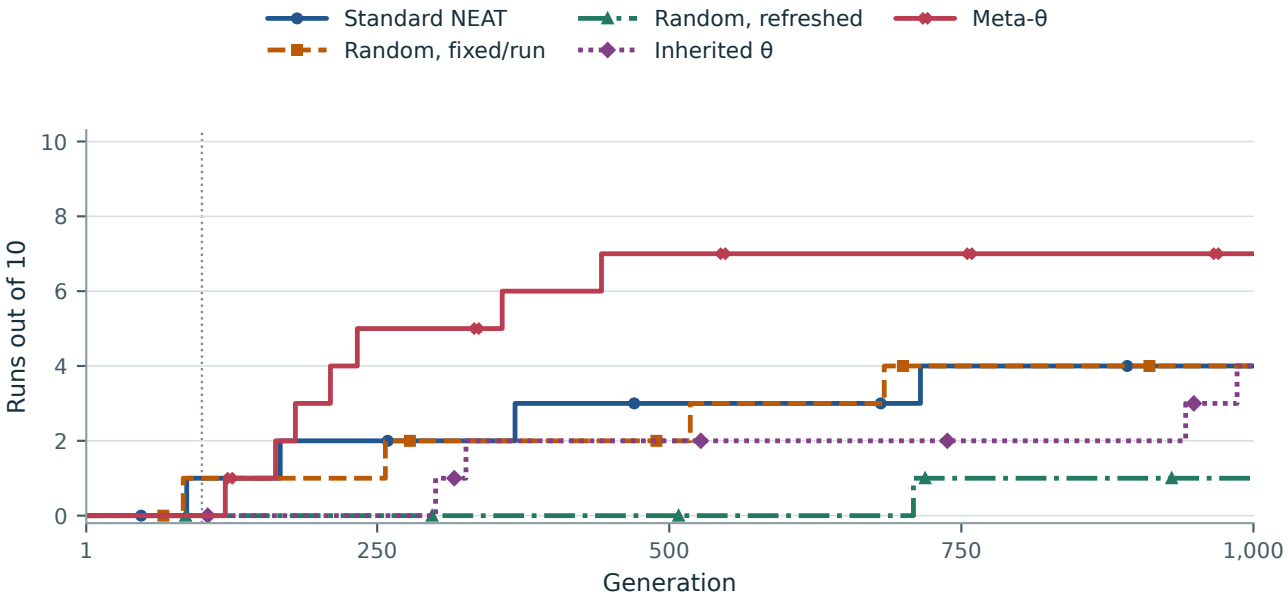
01 / RESULT

Beyond the 0.75 plateau

Five NEAT variants • Seeds 1–10 in each variant • 120 networks per generation

At generation 100, 2 of 50 runs had found a network with fitness above 0.80. By generation 1,000, 20 had done so: 18 additional runs escaped during the extension. At the higher budget, 18 runs reached fitness above 0.95 and 13 training-selected champions classified all 64 unseen cases correctly.



Outcomes at 100 and 1,000 generations

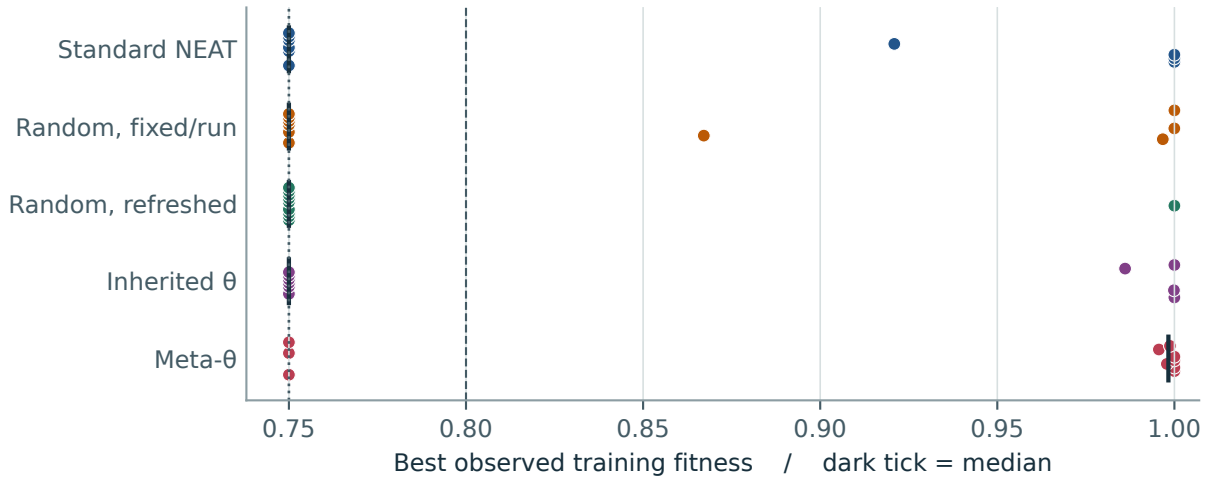
Variant	F > .80 at 100	At 1,000	Best F*	Test acc.*
Standard NEAT	1/10	4/10	0.7500	75.0%
Random, fixed/run	1/10	4/10	0.7500	75.0%
Random, refreshed	0/10	1/10	0.7500	75.0%
Inherited θ	0/10	4/10	0.7500	75.0%
Meta-θ	0/10	7/10	0.9983	100.0%

* Best F and held-out accuracy are medians across the ten runs at generation 1,000. F > .80 counts any strict crossing by the stated generation, including runs that later lost the network.

02 / FITNESS AND GENERALIZATION

What the median curve hides

Each marker on the upper plot is one seed’s best network by generation 1,000.



Thresholds and held-out performance

Variant	> .75[1]	> .80	> .95	64/64[2]	Pop. F[3]
Standard NEAT	4	4	3	2	0.705
Random, fixed/run	4	4	3	0	0.635
Random, refreshed	1	1	1	1	0.664
Inherited θ	4	4	4	4	0.697
Meta-θ	7	7	7	6	0.767

[1] Here the count uses F > 0.750001 to exclude rounding-level differences. [2] Champions with 64/64 correct on the held-out grid. [3] Median of the final within-population mean fitness. All counts are out of ten.

Across all 50 runs, 20 strictly exceeded 0.75, 20 exceeded 0.750001, and 20 exceeded 0.80. The last criterion asks for a reduction of more than 20% in squared error relative to the plateau. Fitness and classification accuracy answer different questions: confident mistakes can offset correct classifications.

03 / SEED TIMING

When did each run escape?

First generation with best training fitness strictly above 0.80, by variant and seed.

	Seed									
	1	2	3	4	5	6	7	8	9	10
Standard NEAT	—	87	715	167	—	—	368	—	—	—
Random, fixed/run	—	518	684	—	84	—	—	—	—	257
Random, refreshed	—	—	—	—	709	—	—	—	—	—
Inherited θ	326	—	942	—	—	—	—	—	986	300
Meta-θ	—	357	163	210	120	442	—	180	233	—

A dash means the run had not crossed by generation 1,000. It does not imply the run would never succeed. All times use generation 1 for the initial population. Each seed gets the same budget; there is no early stopping.

Reading the distribution

The first new escape after the original cutoff was Meta-θ, seed 5, at generation 120. The latest observed crossing was Inherited θ, seed 9, at generation 986. A median fitness curve can stay at 0.75 even when multiple individual runs have learned a better predictor.

What this establishes

Meta-θ recorded the highest escape count (7/10); standard NEAT crossed in 4/10. The 100- and 1,000-generation observations are paired within the same trajectories. Ten seeds per variant leave considerable uncertainty about the ranking on new seeds or a different task.

04 / METHODS

How to reproduce the comparison

The only change to the evolutionary experiment was its duration.

Task and scoring

One calibration label b at the reference reading +1 identifies a sensor's wiring. For a query x, the target is b when $x > 0$ and $1 - b$ when $x < 0$. Both wirings are evaluated on 16 signed readings (odd magnitudes divided by 16). Training fitness is $1 - \text{MSE}$ over these 32 cases; a constant prediction of 0.5 scores 0.75.

We freeze the highest-scoring network and its evaluated control inputs before any rewrite. It is tested at 32 new signed readings (odd magnitudes divided by 32) for both wirings. Accuracy thresholds the output at 0.5. No test result influences evolution; pairing each test case with both possible displayed cues gave exactly 50% accuracy for every champion.

Five conditions, one evolutionary budget

All use the authors' fully connected feedforward start with no hidden nodes: two task inputs and eight normalized controls; one task output and eight control outputs. Standard NEAT fixes Table 2 settings. Random fixed/run draws one fixed vector per run, following the repository. Random refreshed redraws per genome after evaluation, inherited θ evolves the vector, and Meta-θ replaces it with the averaged control outputs.

Population size is 120. The same seeds (1–10), fork, configuration, and random streams were used as in the 100-generation comparison. All 50 runs reproduced their entire earlier 100-generation histories exactly, including population means. In total: 6,000,000 training genome evaluations and 192,000,000 training case evaluations. The earlier per-genome sensitivity check is not one of the five variants.

Code and data

Endogenous NEAT / 12dcffd7ed95

Pinned NEAT-Python fork / 7d9acae0a1e5

The archive includes the runner, task scorer, raw histories, frozen champions, results tables, 100-generation reference records, plotting code and exact commands. The saved generation and seed data give unrounded values and threshold-crossing times.

05 / ARCHITECTURE

How small a network solves the task?

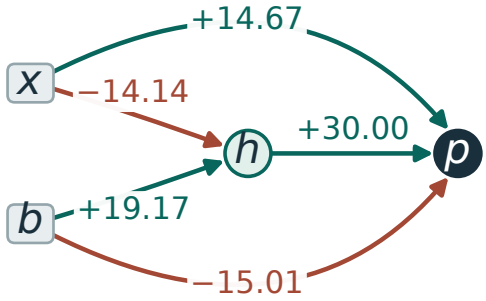
Saved network weights and forward passes from two training-selected champions.

All 32 training examples



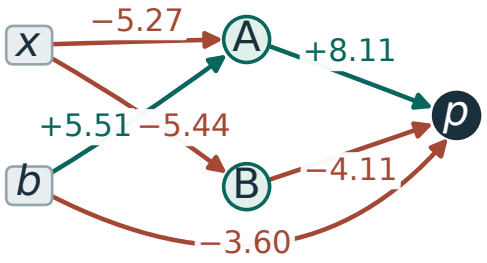
The evolved prediction circuits

Standard NEAT, seed 4



Biases h −0.09 p −15.00

Meta-θ, seed 2



Biases A +0.06 B +0.17 p −1.98

Both champions classified 32/32 training and 64/64 unseen cases correctly. Arrows carry the recorded weights. The constant control inputs are absorbed into the displayed biases. Each circle applies the sigmoid with slope factor five.

Four training cases replayed through Meta-θ, seed 2

Reading	Cue	Target	Unit A	Unit B	P(light)
−9/16	0	1	1.0000	1.0000	0.999957
−9/16	1	0	1.0000	1.0000	0.000361
+9/16	0	0	0.0000	0.0000	0.000050
+9/16	1	1	1.0000	0.0000	0.999997